

Partial Fiscal-Space Transfer and European Bond Pricing

Dario Moni Bidin

1 Introduction

The euro-area sovereign debt crisis and the COVID-19 pandemic revived the debate on common European debt. While a common bond can reduce market fragmentation, provide a safe asset, and lower borrowing costs for fiscally stressed countries, it also raises the concern that weaker sensitivity of national funding costs to national fiscal positions may erode fiscal discipline, and that the major gains can be obtained only by the less fiscally virtuous members.

In light of this debate, this paper extends the fiscal-limit framework of Pallara and Renne (2023), which prices Eurobonds by linking sovereign default risk to fiscal space, the distance between a country's fiscal limit and its public debt. Their analysis compares two polar structures: under several and joint guarantees (SJG), fiscal resources are pooled and the common bond benefits from diversification; under several but not joint guarantees (SNJG), each country is responsible only for its own share, so the common bond behaves like a weighted portfolio of national bonds.

The main contribution of this work is a parameter $\theta \in [0, 1]$ governing the fraction of national fiscal space transferred to a European issuer, spanning the range between purely national issuance and full fiscal mutualisation. This continuous parametrisation goes beyond the binary SJG/SNJG comparison: it identifies a minimum effective threshold below which diversification gains collapse, shows that the transition to meaningful risk-sharing is non-linear, and provides a tractable basis for evaluating institutionally realistic designs where fiscal capacity is introduced gradually. A further result concerns the sign reversal of the SJG–SNJG spread differential when aggregate fiscal space turns sufficiently negative: below a critical threshold, pooling raises rather than lowers the common bond yield, bounding the range of θ for which mutualisation is beneficial. Redistribution is essential to share gains broadly; GDP-weighted, equal-gain, or protected-funding-cost schemes preserve some market discipline while avoiding concentration of benefits in high-spread countries.

2 A Fiscal-Space Model with European Fiscal Capacity

The model builds on the fiscal-limit approach to sovereign credit risk, whose central insight is that a country's default risk depends not on the absolute level of its debt, but on its distance from the fiscal limit, defined as the maximum debt sustainable given the expected path of future primary surpluses and the government's capacity for fiscal adjustment. In the framework developed by Pallara and Renne (2023), fiscal limits are not directly observable but can be recovered from sovereign spreads through

the model's pricing equations, which establishes a disciplined mapping between market prices and fiscal fundamentals. The estimation exploits their inversion technique (adapted from Chen and Scott, 1993), whereby fiscal limits are recovered by inverting the pricing equations at each date, with the remaining parameters estimated by maximum likelihood.

Their main application compares two polar guarantee structures for a common bond. Under several and joint guarantees (SJG), fiscal resources are fully pooled across member states, generating diversification benefits when national fiscal risks are imperfectly correlated, but potentially weakening market discipline for individual countries. Under several but not joint guarantees (SNJG), each country remains responsible only for its own share, so the common bond reduces to a weighted portfolio of national claims with no genuine risk sharing.

The present work retains the fiscal-limit logic but introduces a European issuer backed by only a fraction θ of national fiscal space, yielding a model that is intermediate between these two polar cases and suitable for analysing gradual steps toward European fiscal integration.

2.1 National fiscal space and default risk

Consider a monetary union composed of N member countries, indexed by $i \in \{1, \dots, N\}$, each of which issues sovereign debt and is subject to its own fiscal constraints. At each point in time t , country i is characterised by two state variables: its debt-to-GDP ratio $d_{i,t} \geq 0$, which measures the stock of outstanding public liabilities as a fraction of national output, and its fiscal-limit-to-GDP ratio $\ell_{i,t}$, which captures the maximum level of debt that the country can sustainably carry given its expected future path of primary surpluses and its capacity to adjust fiscal policy in response to adverse shocks. The fiscal space of country i at time t is then defined as the difference between these two quantities:

$$FS_{i,t} = \ell_{i,t} - d_{i,t}. \quad (1)$$

When $FS_{i,t} > 0$, the country's debt lies below its fiscal limit, meaning that there exists positive capacity to absorb further borrowing without triggering a solvency concern. Conversely, when $FS_{i,t} < 0$, public debt has exceeded the fiscal limit, signalling that the country can no longer credibly service its obligations from expected future fiscal revenues, and default risk therefore becomes strictly positive.

The national default intensity $\lambda_{i,t}$, which governs the instantaneous probability of a sovereign credit event, is assumed to be a function solely of the extent to which debt violates the fiscal limit, and is zero whenever fiscal space is non-negative:

$$\lambda_{i,t} = \gamma_i \max\{0, d_{i,t} - \ell_{i,t}\}, \quad (2)$$

where the parameter $\gamma_i > 0$ is a country-specific sensitivity coefficient that measures how rapidly default risk escalates as debt moves beyond the fiscal limit. A higher value of γ_i implies that even a small violation of the fiscal limit translates into a large increase in the probability of default, reflecting either weaker institutional credibility or a narrower set of available fiscal adjustment tools. The corresponding one-period probability of default is obtained from the intensity via the standard Poisson

formula:

$$PD_{i,t} = 1 - \exp(-\lambda_{i,t}). \quad (3)$$

This formulation implies that $PD_{i,t} = 0$ whenever $d_{i,t} \leq \ell_{i,t}$, and rises continuously and convexly as the debt-to-GDP ratio exceeds the fiscal limit. The overall structure connects the pricing of sovereign debt directly to observable and inferable fiscal state variables, allowing default probabilities to evolve endogenously as fiscal conditions change over time.

2.2 Partial transfer of fiscal space

The central institutional innovation introduced in this paper is the notion of a partial transfer of national fiscal space to a supranational European issuer, for example the European Commission, operating on behalf of the member states. Rather than assuming either full fiscal union, in which all resources are pooled, or complete fiscal separateness, in which no risk sharing takes place, the model allows for an intermediate arrangement in which each country i transfers a fraction $\theta \in [0, 1]$ of its fiscal space to the European level, while retaining the remainder at the national level. Formally, the fiscal space transferred by country i to the European issuer at time t is

$$FS_{EU,t}(\theta) = \theta (\ell_{i,t} - d_{i,t}). \quad (4)$$

The parameter θ therefore plays a dual role in the model: it simultaneously governs the strength of the backing available to the European bond and the residual fiscal capacity that national governments retain for their own debt management. This captures a fundamental institutional trade-off where a more credible and better-backed European issuer requires reallocating a larger share of fiscal capacity away from the national level. Crucially, countries do not transfer debt directly to the European issuer; rather, they transfer a share of the fiscal space that backs their debt, so the European bond is priced according to the distribution of fiscal positions across member states, not merely by the level of aggregate public debt.

The parameter θ is treated as a policy-determined constant set by treaty or intergovernmental agreement. Endogenising it would require embedding θ in a strategic game among N governments with asymmetric incentives, shifting the focus from bond pricing to political-economy equilibria. A natural extension would model θ as the Nash outcome of a bargaining problem in which each government trades off the pricing gains from pooling against the loss of national fiscal capacity and the associated reduction in market discipline; the redistribution weights $\omega_{G,j}$ could serve as the transfer instrument making participation individually rational. This extension is left for future research.

Let $\omega_i \geq 0$ denote the participation weight of country i in the common issuer, normalised so that $\sum_{i=1}^N \omega_i = 1$. These weights can be interpreted variously as GDP shares. The aggregate fiscal backing available to the European issuer at time t , after the partial transfer of national fiscal spaces, is

$$F_t^{EU}(\theta) = \theta \sum_{i=1}^N \omega_i (\ell_{i,t} - d_{i,t}). \quad (5)$$

This quantity represents the weighted average fiscal space of the member states, scaled by the transfer fraction θ , and constitutes the total resource pool from which the European issuer can meet its obligations. If the European issuer has outstanding common debt b_t^{EU} , measured as a fraction of euro-area GDP, then its net fiscal space and default intensity are, respectively,

$$FS_t^{EU}(\theta) = \theta \sum_{i=1}^N \omega_i (\ell_{i,t} - d_{i,t}) - b_t^{EU}, \quad (6)$$

$$\lambda_t^{EU}(\theta) = \gamma_{EU} \max \left\{ 0, b_t^{EU} - \theta \sum_{i=1}^N \omega_i (\ell_{i,t} - d_{i,t}) \right\}, \quad (7)$$

where $\gamma_{EU} > 0$ is the sensitivity of European default risk to a shortfall in fiscal backing. Following Pallara and Renne (2023), γ_{EU} is set equal to the participation-weighted average of national sensitivities:

$$\gamma_{EU} = \sum_{i=1}^N \omega_i \gamma_i. \quad (8)$$

This choice is motivated by parsimony: introducing an additional free parameter would require identifying it separately from θ , which is not feasible given the counterfactual nature of the European bond. The weighted average constitutes a conservative upper bound on the true sensitivity, since diversification dampens the transmission of individual fiscal deteriorations to the common issuer; the actual default intensity would therefore rise more slowly than implied by (7). The European issuer becomes risky when its outstanding liabilities exceed the transferred fiscal backing, precisely mirroring the logic applied at the national level. A larger value of θ reduces European default risk when aggregate fiscal space is positive, because it increases the pool of backing resources available to the common issuer. However, this comes at the direct cost of compressing the fiscal capacity that remains available to national governments for their own borrowing needs.

2.3 Pricing of the European bond

The pricing of the European bond follows the same reduced-form affine framework employed in Pallara and Renne (2023) for national sovereign bonds. Let $M_{t,t+h}$ denote the stochastic discount factor, which encodes both the risk-free term structure and the compensation that investors require for bearing aggregate risk. Let $RR^{EU} \in [0, 1]$ denote the recovery rate on European debt, i.e. the fraction of face value that bondholders receive in the event of default, and let $B_{t,h}$ denote the time- t price of a risk-free zero-coupon bond maturing at $t + h$. The price of a risky European zero-coupon bond with maturity h , as a function of the transfer parameter θ , is given by

$$P_{t,h}^{EU}(\theta) = (1 - RR^{EU}) E_t [M_{t,t+h} (1 - D_{t+h}^{EU}(\theta))] + RR^{EU} B_{t,h}, \quad (9)$$

where $D_{t+h}^{EU}(\theta) \in \{0, 1\}$ is an indicator that equals one if the European issuer has defaulted by date $t + h$ and zero otherwise. This expression follows directly from the Recovery of Treasury convention of Duffie and Singleton (1999) as applied in Pallara and Renne (2023), extended here to the European

issuer with default intensity (7). The first term captures the risk-adjusted expected payoff in the no-default state, discounted at the stochastic discount factor, while the second term captures the recovery value obtained even in default. The corresponding yield and credit spread of the European bond are

$$i_{t,h}^{EU}(\theta) = -\frac{1}{h} \log P_{t,h}^{EU}(\theta), \quad s_{t,h}^{EU}(\theta) = i_{t,h}^{EU}(\theta) - i_{t,h}^0, \quad (10)$$

where $i_{t,h}^0$ is the yield on the risk-free bond of the same maturity, so that $s_{t,h}^{EU}(\theta)$ measures the excess yield demanded by investors to hold the European bond rather than a default-free alternative.

2.4 Relation with SJG and SNJG bonds

The model introduced in this paper can be understood as occupying an intermediate position between the two polar guarantee structures studied by Pallara and Renne (2023), namely the SJG and SNJG. A clear characterisation of these two benchmarks helps to locate the present framework in the theoretical space and to clarify what is gained by the partial-transfer approach.

In the SJG benchmark, both debt and fiscal limits are fully pooled across all participating countries, weighted by their participation shares:

$$d_t^{SJG} = \sum_i \omega_i d_{i,t}, \quad \ell_t^{SJG} = \sum_i \omega_i \ell_{i,t}. \quad (11)$$

The default intensity of the common issuer then depends solely on the aggregate pooled fiscal position $\ell_t^{SJG} - d_t^{SJG}$, which is a GDP-weighted average of national fiscal spaces. This full pooling generates diversification benefits: a deterioration in the fiscal position of one country does not necessarily imply a deterioration of the aggregate position if other member states have sufficient fiscal space to compensate.

In the SNJG benchmark, by contrast, there is no joint fiscal backing whatsoever. Each country remains solely responsible for its own share of the common instrument, and the price of the SNJG bond is therefore simply a participation-weighted average of national bond prices:

$$P_{t,h}^{SNJG} = \sum_i \omega_i P_{i,t,h}^N, \quad (12)$$

where $P_{t,h}^N$ is the price of country i 's national sovereign bond at maturity h . This structure does not generate spread compression beyond what is already implied by the individual national spreads, because it merely repackages national claims into a common instrument without creating any genuine joint liability or pooling of fiscal capacity.

The European fiscal-space bond introduced in this paper lies strictly between these two limiting cases. It is more integrated than SNJG because the European issuer is backed by a genuine common pool of fiscal space, so that the default risk of the European bond reflects the joint fiscal position of the member states rather than a simple linear combination of individual national risks. At the same time, it falls short of full SJG integration because the transfer of fiscal space is only partial: only the fraction θ of national fiscal capacity is contributed to the common pool, while the remaining fraction $(1 - \theta)$ is

retained at the national level. This parameterisation makes the model particularly suitable for studying institutional designs in which European fiscal capacity is created gradually and incrementally, rather than through an abrupt and complete transition to full fiscal union.

2.5 Debt and fiscal-limit dynamics

To implement the model quantitatively and conduct simulation and counterfactual experiments, debt-to-GDP ratios and fiscal limits are assumed to follow persistent reduced-form autoregressive processes. Specifically, for each country i , the dynamics of $d_{i,t}$ and $\ell_{i,t}$ are governed by

$$d_{i,t+1} = (1 - \rho_d) \bar{d}_i + \rho_d d_{i,t} + \sigma_{d,i} \varepsilon_{i,t+1}^d, \quad \ell_{i,t+1} = (1 - \rho_\ell) \bar{\ell}_i + \rho_\ell \ell_{i,t} + \sigma_{\ell,i} \varepsilon_{i,t+1}^\ell. \quad (13)$$

Here, $\bar{d}_i$ and $\bar{\ell}_i$ denote the country-specific unconditional means of the debt ratio and the fiscal limit, respectively, toward which each process reverts at rates $\rho_d \in (0, 1)$ and $\rho_\ell \in (0, 1)$; $\sigma_{d,i}$ and $\sigma_{\ell,i}$ are country-specific volatility parameters; and $\varepsilon_{i,t+1}^d$ and $\varepsilon_{i,t+1}^\ell$ are the innovation shocks. These shocks are decomposed into a common euro-area component, shared by all member states, and an idiosyncratic country-specific component that affects only the individual country. This two-factor structure is important because the welfare gains from common issuance depend critically on the cross-country correlation of fiscal risks: if national fiscal shocks are perfectly correlated, pooling provides no diversification benefit, since all countries deteriorate simultaneously and the aggregate fiscal position moves in lockstep with each individual one; if, on the other hand, shocks are imperfectly correlated, a common fiscal-space bond can meaningfully reduce the probability that the European issuer exhausts its transferred backing, because adverse developments in one country are likely to coincide with more favourable conditions elsewhere.

3 Model Estimates and Policy Implications

3.1 Model fit and fiscal limits

The model captures the principal features of sovereign risk dynamics across the estimation sample. Sovereign CDS spreads are reproduced closely at the 3, 5 and 10-year maturities for all six countries, including both the sharp widening during the euro-area sovereign debt crisis and the subsequent compression (Figure 1). The cross-sectional ordering of credit risk is also well recovered: Italy and Spain exhibit the largest spread increases, Germany and the Netherlands remain at the lower end of the distribution throughout, and France and Belgium occupy an intermediate position. A partial exception arises for France at the 10-year maturity in the post-2015 period, where the model underestimates spreads. This discrepancy is consistent with the ECB's forward guidance era, during which spread compression was partly driven by monetary policy expectations rather than by improvements in fiscal fundamentals; a purely fiscal-limit model therefore attributes too much safety to France relative to what market prices implied. The inferred French fiscal limit should accordingly be interpreted with caution for the later part of the sample.

The estimated fiscal limits are broadly consistent with values reported in the related literature, notwithstanding the inherent model-dependence of such estimates (Table 1), and the cross-sectional ranking of fiscal space mirrors that of sovereign spreads, supporting the interpretation of fiscal space as a summary measure of creditworthiness (Figure 3). The high persistence of both debt and fiscal-limit dynamics, reported in Table 2, implies that fiscal conditions affect pricing not only at short maturities but also across the medium and long ends of the yield curve. Because the European issuer is backed by the weighted pool of transferred fiscal space, the price of the European bond depends not only on the average fiscal stance of the monetary union but also on the dispersion of fiscal positions across participating countries: fiscally stronger members contribute more insurance to the common pool, while weaker ones benefit disproportionately from the pooling arrangement.

3.2 European bond pricing

Partial fiscal-space pooling generates meaningful spread compression relative to a portfolio of national claims, especially during periods of cross-country fiscal divergence. The European fiscal-space bond spread lies strictly below the participation-weighted average of national sovereign spreads throughout the sample, with the gap being most pronounced during the sovereign debt crisis of 2011–2012 (Figure 2). This result reflects the same economic force identified by Pallara and Renne (2023): a common bond backed by a pooled fiscal resource benefits from diversification when national fiscal risks are imperfectly correlated, whereas an SNJG-style instrument remains close to a weighted average of national risks and generates no such compression.

The key distinction relative to Pallara and Renne is that the pooling mechanism here is partial and governed by θ . The European issuer does not receive the full fiscal resources of member states, as in the pure SJG case, but only a fraction θ thereof. The emergence of meaningful spread compression even under partial transfer implies that complete fiscal mutualisation is not a necessary condition for significant risk-sharing benefits: a limited European fiscal capacity is already sufficient to generate spreads below those of a portfolio of national claims. Gains from common issuance are state-dependent and countercyclical, since they are largest precisely when fiscal stress and market fragmentation are highest, and negligible in tranquil periods when national spreads are uniformly low.

The composition of the participating pool is a first-order determinant of European bond pricing. Excluding Germany and the Netherlands from the pool removes the countries with the largest positive fiscal space and therefore weakens the common backing, raising implied spreads appreciably. Excluding Spain and Italy, conversely, eliminates the riskiest fiscal positions and lowers spreads substantially. These counterfactuals confirm that the European bond spread is not a mechanical function of aggregate debt but reflects the fiscal quality and heterogeneity of the participating pool (Figures 6, 7, and 8). Pool composition, including participation weights and the extent of individual contributions, is therefore a key design parameter for any common debt instrument backed by fiscal space.

3.3 Redistribution of gains

Spread compression at the aggregate level does not imply that every member state benefits. Whether a country gains or loses from common issuance depends critically on the redistribution scheme adopted, since without ex-post redistribution the gains are concentrated among high-spread countries while fiscally safer countries may face higher borrowing costs. Under $\theta = 0.30$ in 2011, Italy records a yield gain of 441 basis points relative to its national yield while Germany records a loss of 16 basis points (Panel A of Tables 5–7). This outcome reflects the natural distributional implication of common issuance without redistribution: countries with high national yields capture most of the benefit from spread compression.

Once redistribution is introduced, however, partial fiscal-space pooling can generate Pareto-improving outcomes. When gains are distributed according to GDP weights, all countries obtain positive yield reductions. Under $\theta = 0.30$ in 2011, gains range from 117 basis points for Germany to 146 basis points for Italy (Panel B). A redistribution scheme imposing equal yield gains across all member states delivers approximately 125 basis points of gains during the same crisis episode, with smaller but positive gains in 2008 and 2021 (Panel C). The equal-gain scheme is particularly transparent, since it eliminates cross-country differences in the magnitude of the benefit while preserving a direct link between the redistribution rule and fiscal fundamentals. The countercyclical pattern of gains is confirmed across all redistribution schemes: as sovereign spreads diverge, the value of common issuance rises; as they compress, it declines (Figure 10). The maturity dimension also matters, with gains differing across the 3, 5, and 10-year horizons in a manner consistent with the term structure of sovereign risk.

A further set of schemes imposes that the funding costs of fiscally safer countries are held at their national levels. The results show that protecting Germany's or both Germany's and France's funding costs is feasible without eliminating the gains for high-spread countries, although it concentrates a larger share of the aggregate benefit among fiscally weaker members (Table 8).

3.4 Moral hazard and market discipline

A structural implication of common issuance is that it attenuates the sensitivity of national funding costs to national fiscal behaviour, giving rise to a moral-hazard channel. This attenuation can be seen directly from the default intensity of the European issuer. Differentiating $\lambda_t^{EU}(\theta)$ with respect to country i 's debt yields

$$\frac{\partial \lambda_t^{EU}}{\partial d_{i,t}} = -\theta \omega_i \gamma_{EU} \mathbf{1} \left\{ b_t^{EU} > \theta \sum_j \omega_j F S_{j,t} \right\}, \quad (14)$$

where $\mathbf{1}\{\cdot\}$ denotes the indicator for the default-active region. This sensitivity is proportional to $\theta \omega_i$, which is strictly less than the corresponding national sensitivity γ_i whenever $\theta < 1$ or $\omega_i < 1$. Two implications follow directly. First, the moral-hazard channel is structurally present regardless of θ : common issuance always attenuates the link between national debt and national funding costs to some degree. Notably, this attenuation operates even when θ lies below the minimum effective threshold θ^* identified in Section 3.5: the benefits of partial mutualisation are non-linear in θ , while the incentive

distortion is approximately linear, suggesting that the optimal θ balancing the two margins lies above θ^* but strictly below one. Second, the attenuation is decreasing in the country's participation weight ω_i , so that larger countries internalise a greater share of the fiscal cost they impose on the common pool. Under a GDP-weighted scheme, Germany's weight is approximately three times that of Italy, implying that the incentive distortion is considerably smaller for larger economies.

The stress-test results confirm this mechanism empirically: when Italy or Spain unilaterally increase their debt-to-GDP ratio, the common bond yield reacts far less than the corresponding national yield, regardless of the guarantee structure (Figure 5). This is the moral-hazard channel: fiscal resources are pooled, so an individual country does not fully internalise the effect of its own debt accumulation on the common issuer. Redistribution schemes, however, partially restore market discipline. Under GDP-weighted or equal-yield-gain redistribution, post-redistribution funding costs remain more sensitive to national debt than the common bond yield itself. In fact, as a country's debt increases, its allocated gain from redistribution declines, so its effective borrowing cost retains a link to its own fiscal behaviour.

The moral-hazard channel is also state-dependent; in 2011, when Italy and Spain are close to their fiscal limits, national yields respond steeply to additional debt increases and the gap between national and post-redistribution yields is quantitatively large. In 2021, the response is more muted at low debt levels but steepens as the shock pushes the country toward its fiscal limit. This nonlinearity underscores the importance of redistribution design especially in periods of fiscal stress, when the incentive effects of common issuance are most consequential.

3.5 Sensitivity to θ

The main qualitative results are robust across $\theta \in \{0.05, 0.10, 0.20, 0.30, 0.40\}$, while important nonlinearities emerge at low values of θ . The near-insensitivity of results across the upper range of θ admits a structural interpretation rooted in the default intensity equation (7): the European default intensity is activated only when common debt exceeds the transferred fiscal backing $\theta \sum_i \omega_i F S_{i,t}$. When aggregate fiscal space is sufficiently positive, the pool already sits well above the default threshold at $\theta = 0.20$, and increasing it to 0.40 shifts the threshold further without changing the qualitative pricing regime. Spread compression is therefore determined primarily by the cross-sectional dispersion of fiscal space rather than by the level of θ , once a minimum backing is in place. Under the equal-yield-gain scheme, crisis-period gains amount to approximately 128, 125, and 128 basis points for $\theta = 0.20, 0.30$, and 0.40 respectively, confirming the near-flatness of the gain schedule over this range (Tables 5–7).

At $\theta = 0.05$, the mechanism loses traction sharply (Table 3). The equal-yield-gain scheme delivers only 57 basis points of crisis-period gains in 2011, which is less than half the level at $\theta = 0.20$. The European bond is priced close to the SNJG benchmark, with the SNJG same-funding-cost scheme itself generating crisis-period gains of similar order of magnitude (Panel D). At this level of transfer, the European issuer holds too little transferred fiscal backing to generate meaningful diversification benefits, and the pricing of the common bond reflects primarily the weighted average of national risks rather than a genuinely pooled fiscal position.

Taken together, the four lower cases identify a minimum effective threshold for European fiscal capacity lying between $\theta = 0.05$ and $\theta = 0.20$ in this calibration. The transition from negligible to meaningful risk-sharing is non-linear and concentrated in this interval: gains rise steeply as θ increases from 0.05 to 0.10 to 0.20, then flatten substantially for $\theta \in \{0.20, 0.30, 0.40\}$. A European fiscal capacity calibrated below the minimum threshold generates little pricing benefit regardless of the redistribution rule applied, while one calibrated above it captures most of the diversification gains associated with full fiscal union, with sharply diminishing marginal returns to further increases in θ beyond that point.

The analytical characterisation of θ^* remains an open question for future research. In the present calibration it lies in the interval $(0.05, 0.20)$, but its exact location depends on the distribution of fiscal spaces and the cross-country correlation structure. A natural starting point would be to derive a closed-form condition in the two-country stylised model, exploiting the Gaussian structure of the debt and fiscal-limit processes to express the default-intensity activation condition as an explicit function of θ , aggregate fiscal space, and the cross-country covariance matrix.

4 Conclusion

This paper extends the fiscal-limit framework of Pallara and Renne (2023) by introducing a parameter θ measuring the share of national fiscal space transferred to a European issuer, spanning the range from purely national issuance to full fiscal mutualisation. While they establish the dominance of SJG over SNJG bonds through fiscal diversification, their binary comparison cannot reveal a minimum effective threshold or the diminishing returns that emerge above it.

Partial mutualisation compresses European bond spreads and generates aggregate gains, especially during fiscal stress and cross-country divergence. These gains are not monotone in the fiscal cycle: when aggregate fiscal space turns sufficiently negative, pooling raises rather than lowers the common bond yield, providing a natural upper bound on the beneficial range of θ .

Redistribution is essential. Without it, high-spread countries capture most gains while safer countries may be worse off. GDP-weighted, equal-gain, or protected-funding-cost schemes deliver broadly shared gains while maintaining a link between national fiscal conditions and post-redistribution funding costs, partly restoring market discipline. Larger countries internalise a greater share of the fiscal externality through their higher participation weights, suggesting that GDP-weighted participation is preferable on both efficiency and discipline grounds.

The sensitivity analysis across $\theta \in \{0.05, 0.20, 0.30, 0.40\}$ reveals a non-linearity: spread compression declines sharply at $\theta = 0.05$ but is nearly flat above $\theta = 0.20$, identifying a minimum effective threshold θ^* below which diversification gains collapse. The optimal determination of θ remains an open question, requiring a balance between the decreasing marginal gain from higher pooling and the increasing marginal cost of reduced national fiscal capacity—a trade-off that calls for embedding θ in a model with endogenous fiscal policy. The present results already suggest, however, that limited and rule-based European fiscal capacity, if calibrated above the minimum threshold and paired with transparent redistribution rules, can capture most of the pricing benefits of full fiscal union.

5 Appendix

5.1 Figures

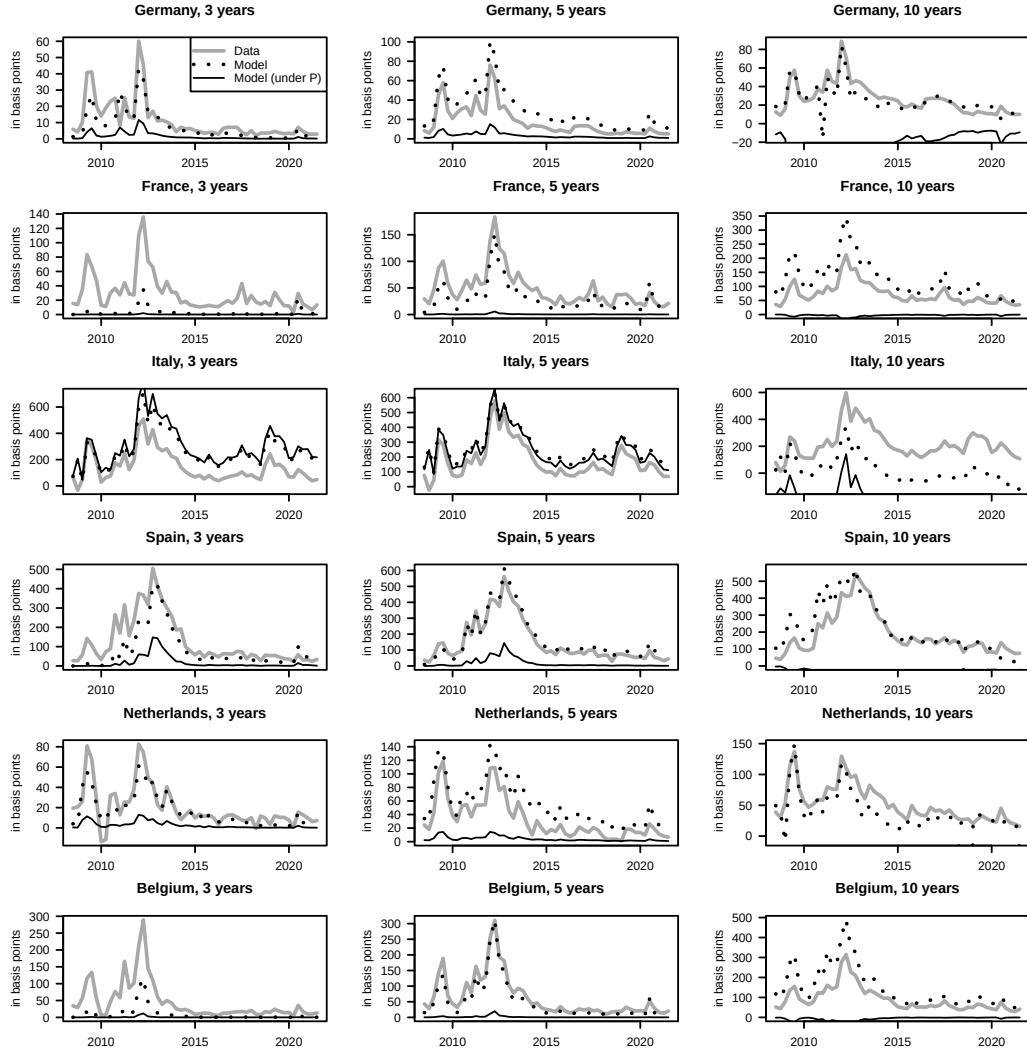


Figure 1: Model fit to sovereign CDS spreads, $\theta = 0.20$.

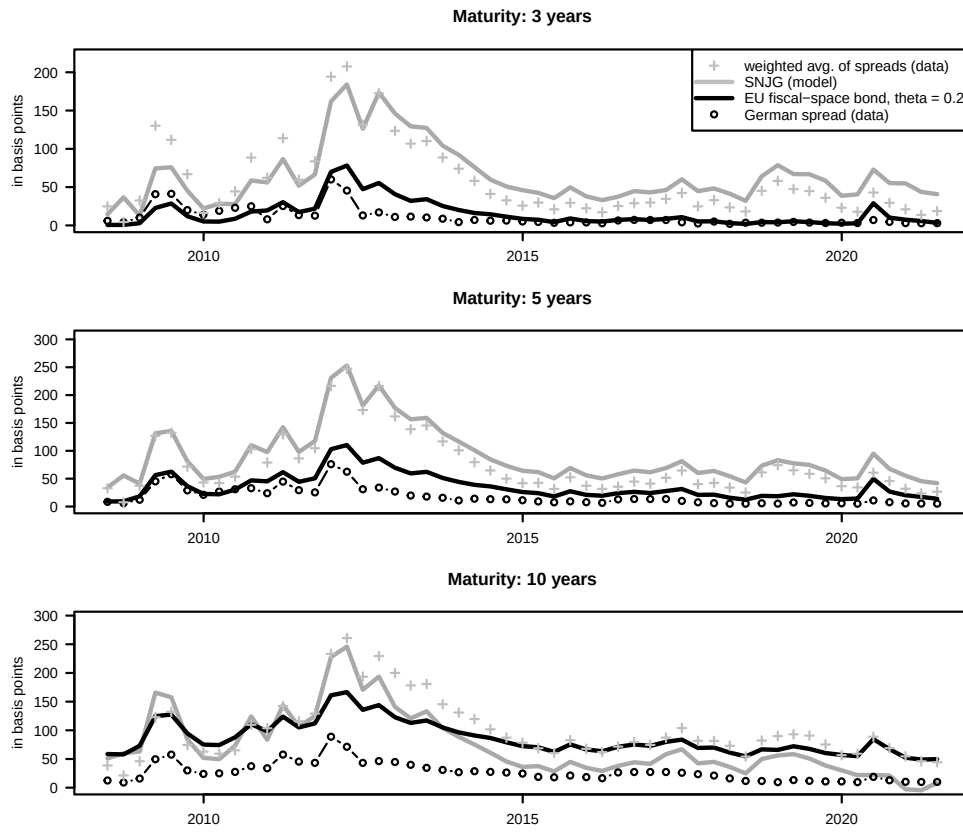


Figure 2: European bond spreads under the baseline calibration.

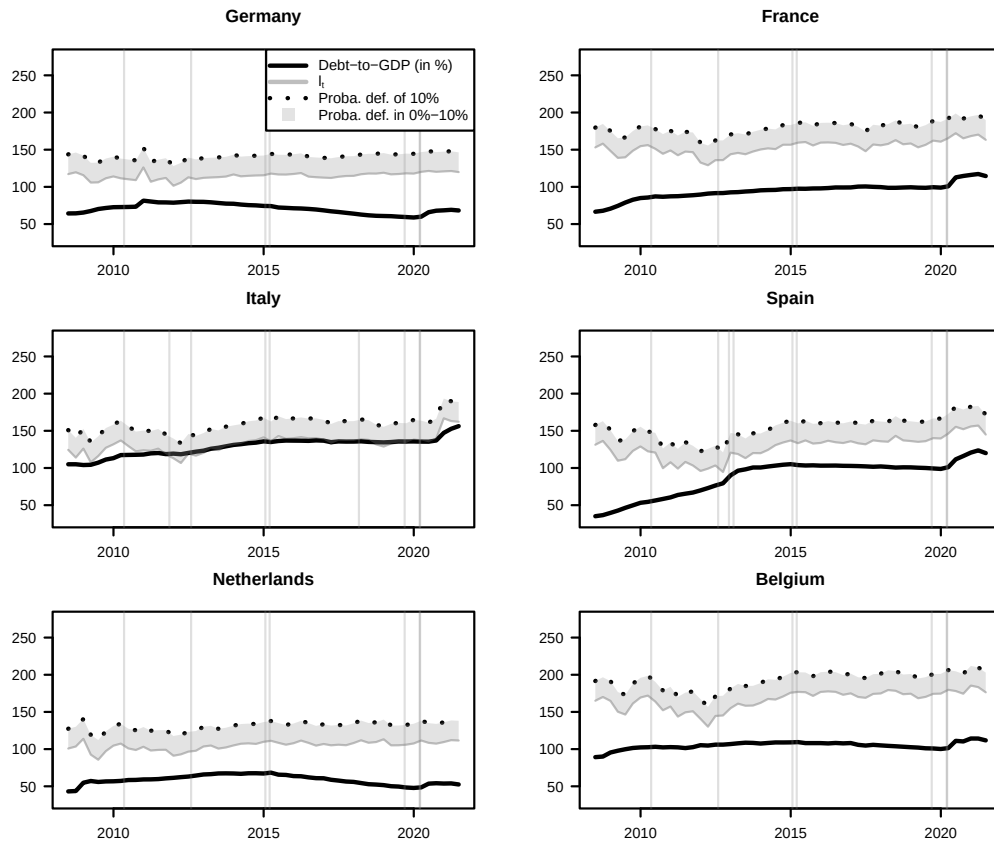


Figure 3: Fiscal limits and debt-to-GDP ratios under the baseline calibration.

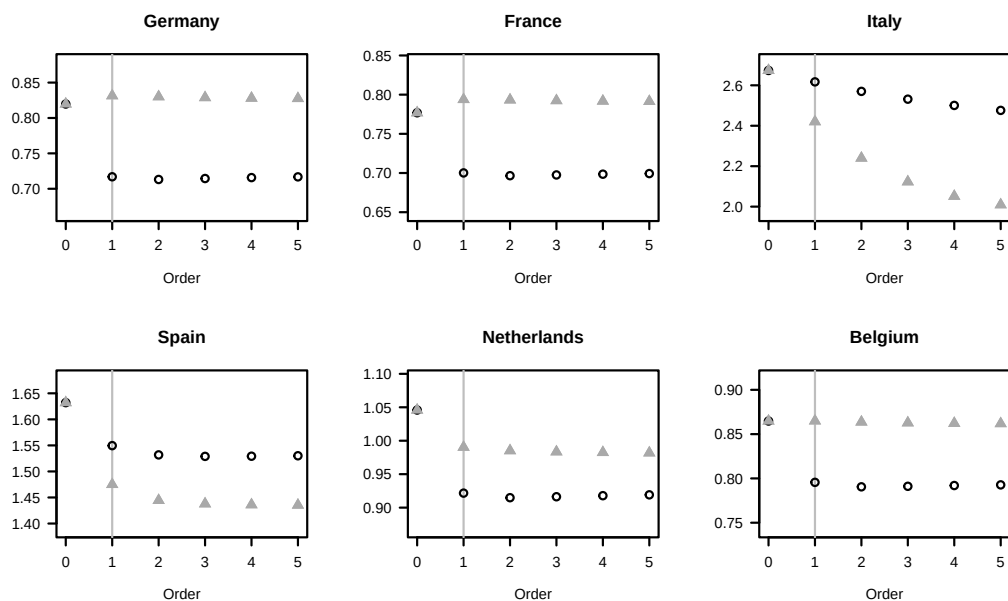


Figure 4: Higher-order approximation diagnostics under the baseline calibration.

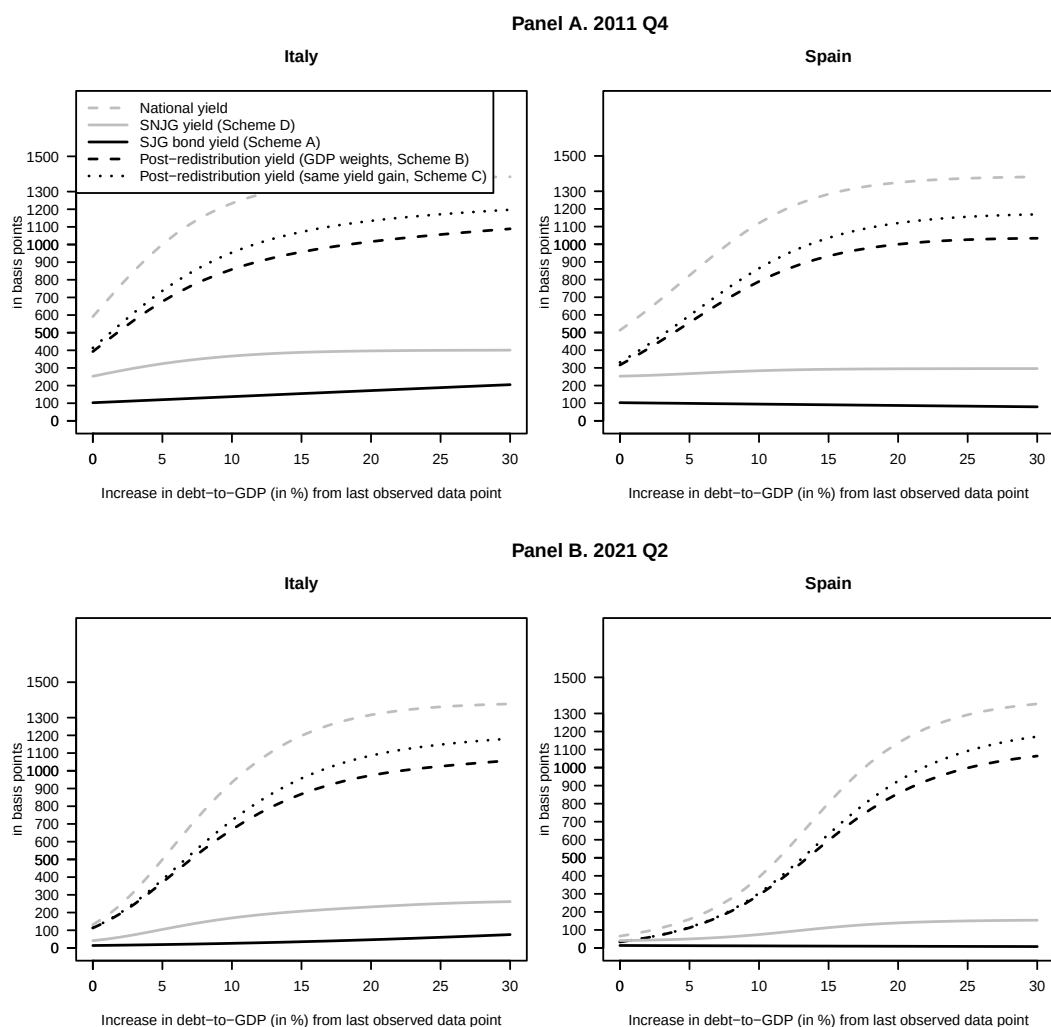


Figure 5: Yield effects of debt-to-GDP stress scenarios for Italy and Spain under the baseline calibration.

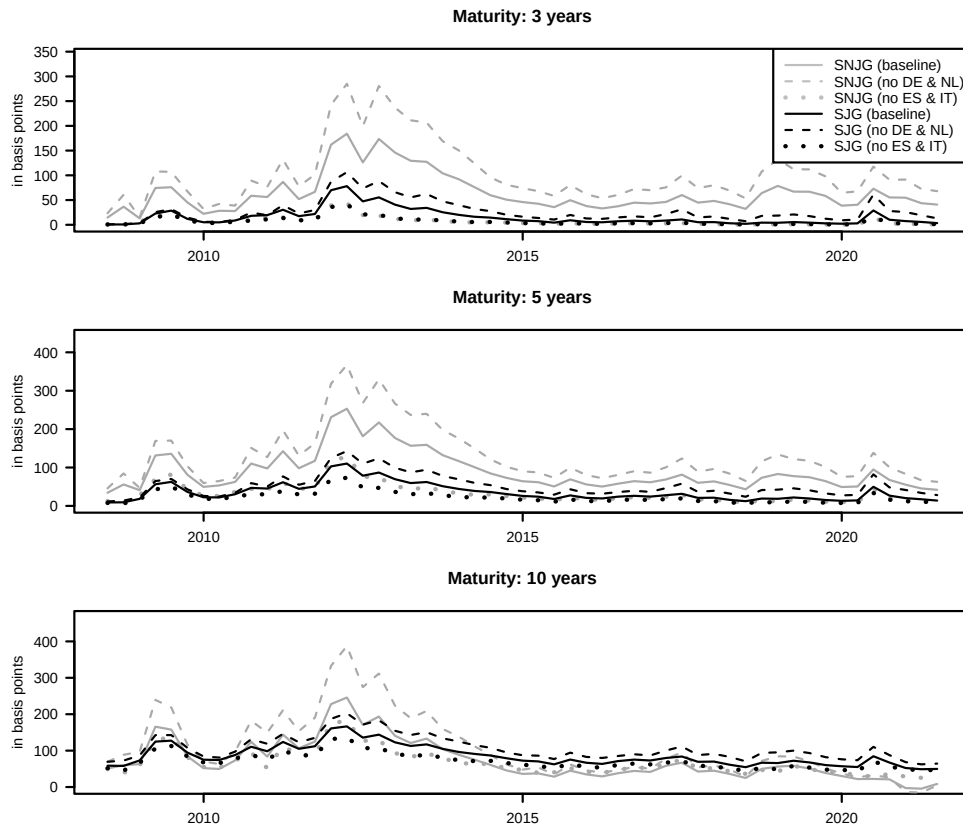


Figure 6: European bond spreads with the complete set of countries under the baseline calibration.

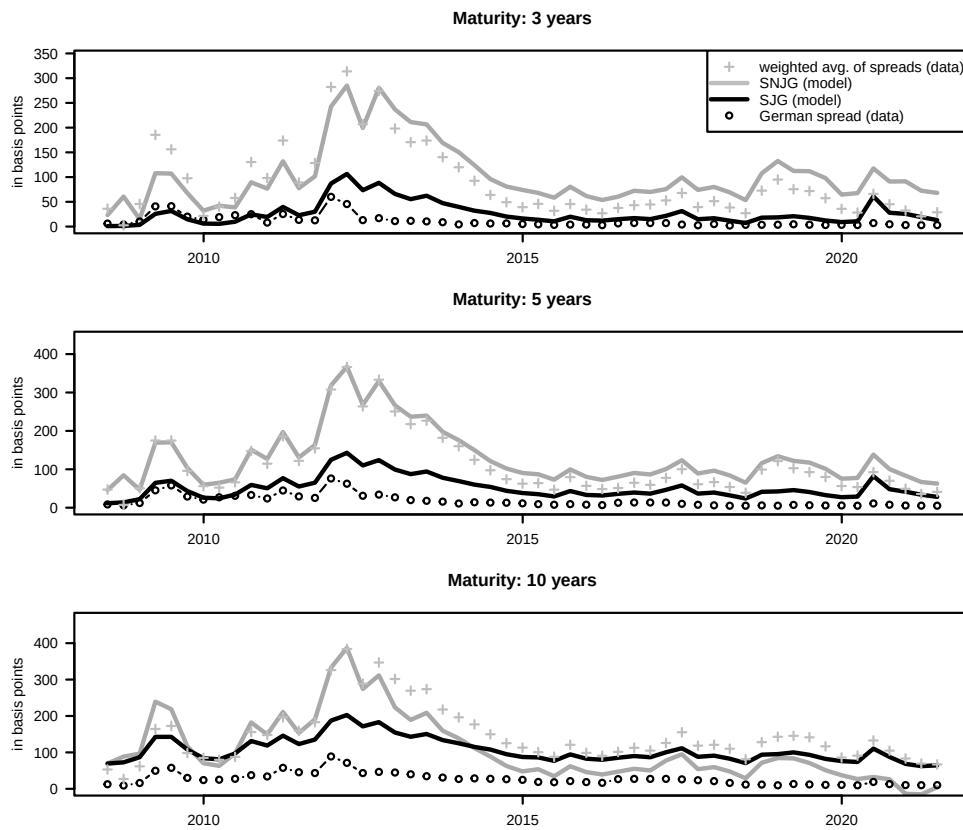


Figure 7: European bond spreads excluding Germany and the Netherlands under the baseline calibration.

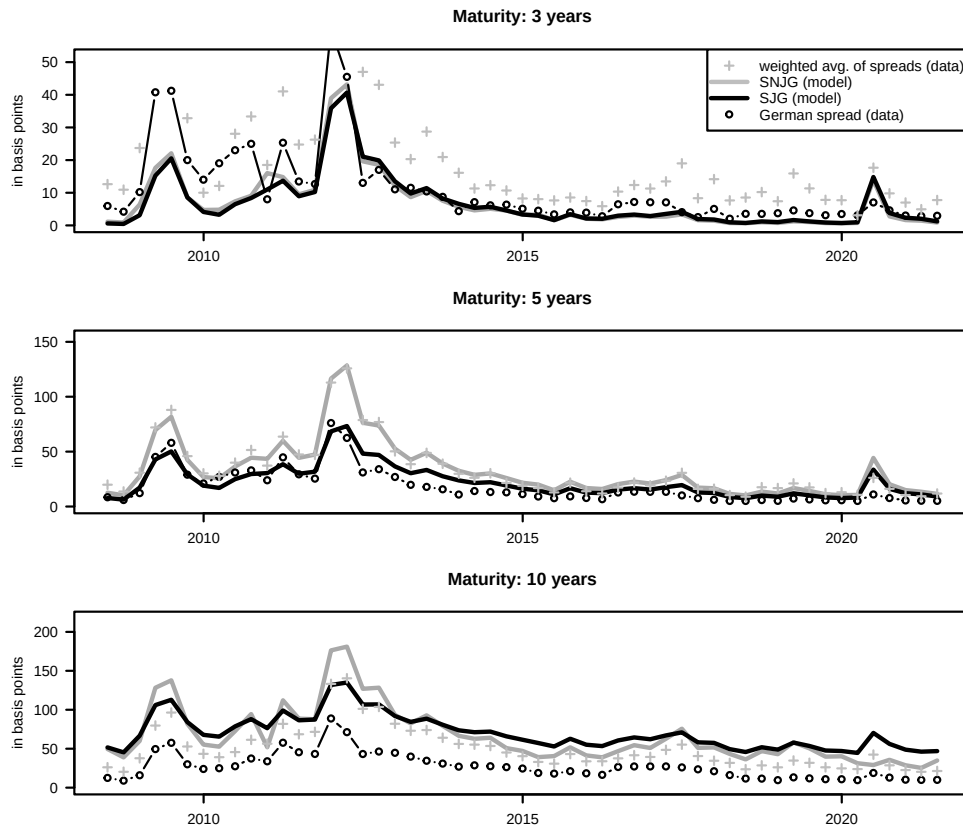


Figure 8: European bond spreads excluding Spain and Italy under the baseline calibration.

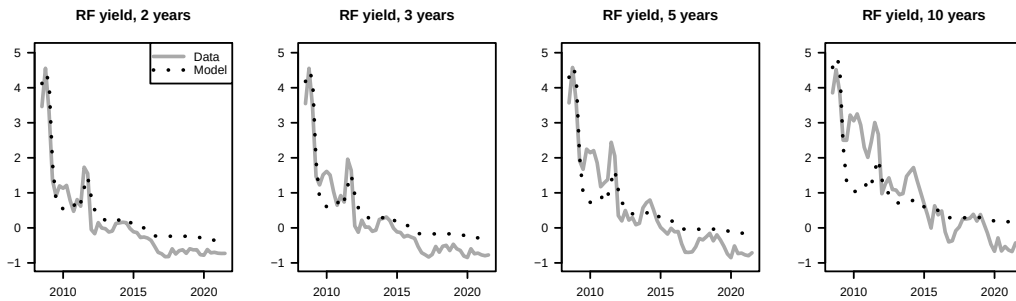


Figure 9: Model fit to risk-free yields under the baseline calibration.



Figure 10: Redistribution scheme with equal yield gains across countries.

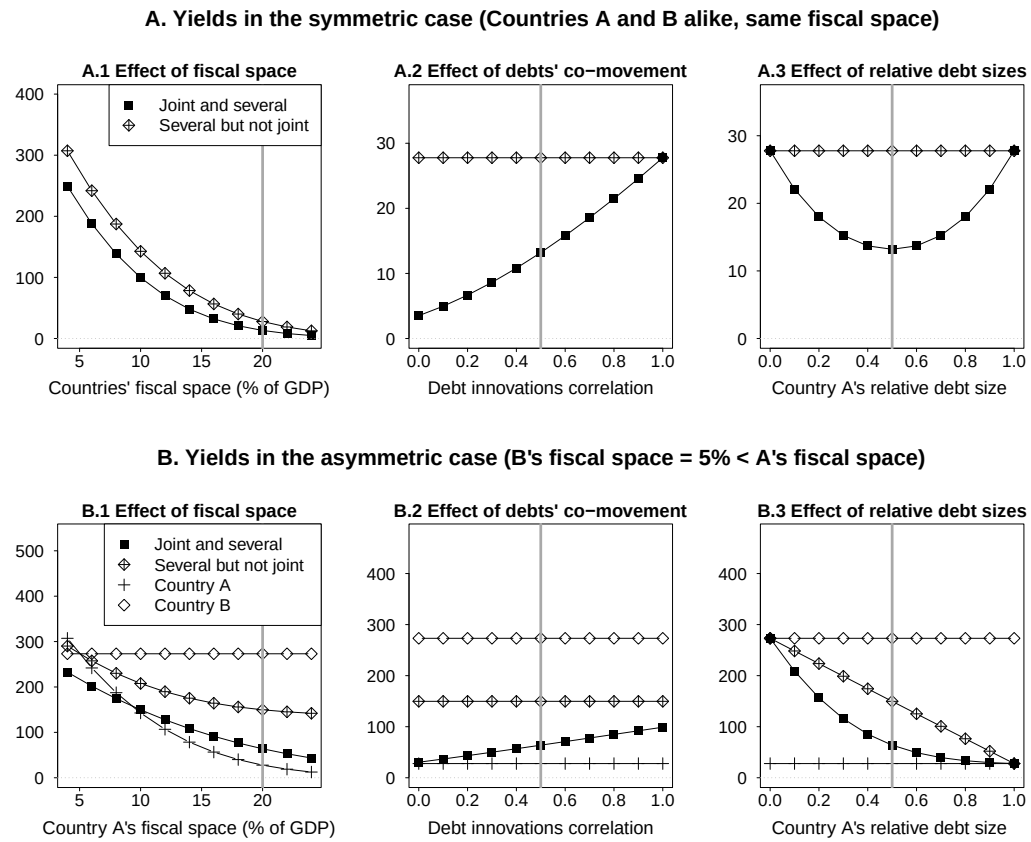


Figure 11: Toy model illustration of joint and several versus several-but-not-joint guarantees.

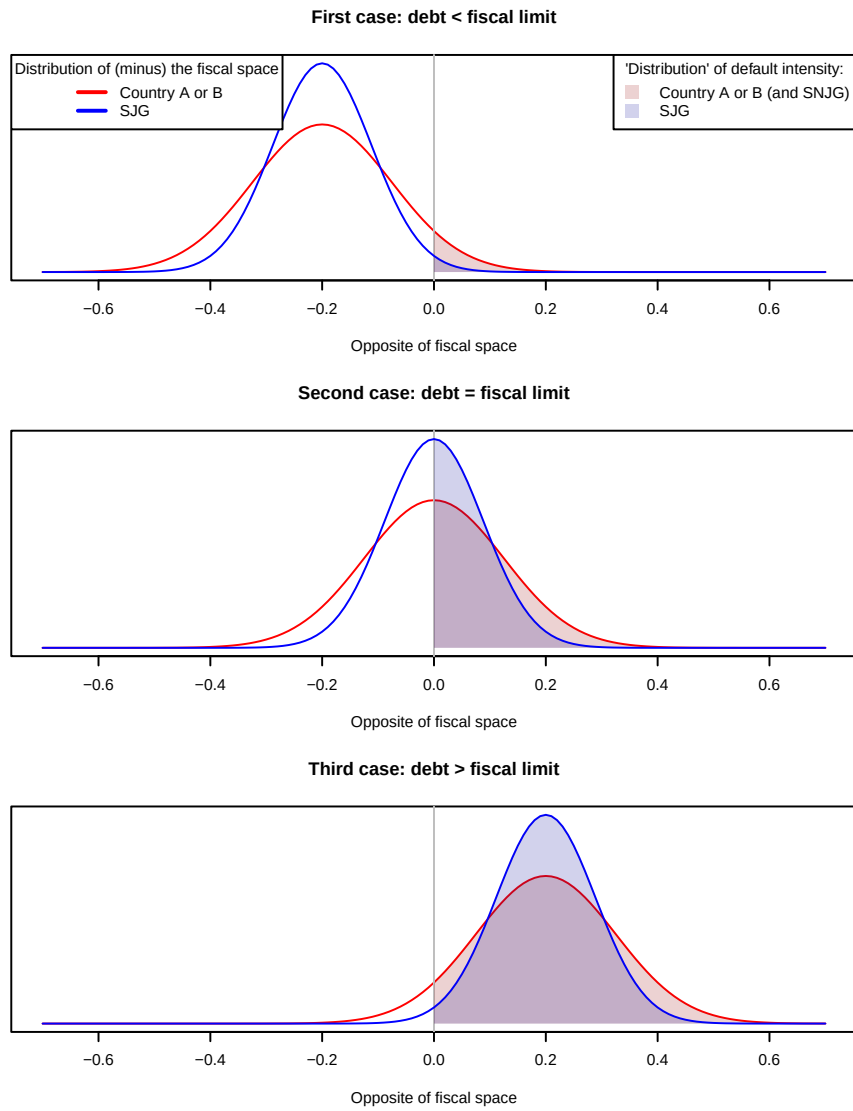


Figure 12: Toy model illustration in terms of default-intensity distributions and fiscal space.

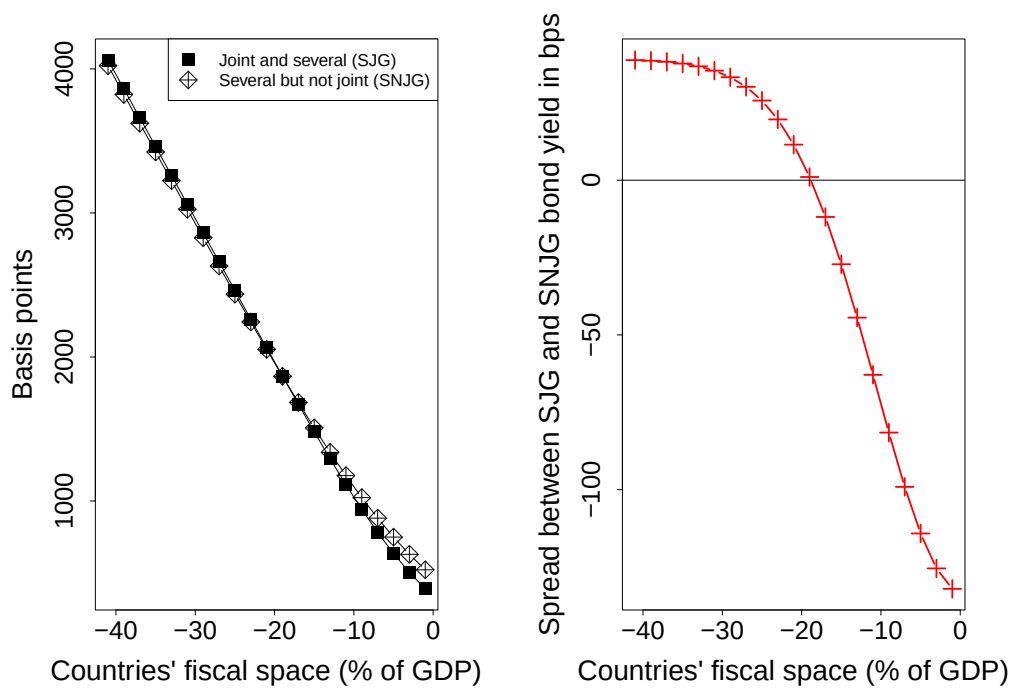


Figure 13: SJG vs. SNJG bond yields as aggregate fiscal space turns negative.

5.2 Tables

Table 1: Fiscal limit estimates from the literature

Ctry	Ghosh et al. (2013)		Collard et al. (2015)					
	Hist.	Proj.	5% MPS	MRR	TVR	CATA	4% MPS	h. MPS
DE	154.1	175.8	130.1	132.3	114.6	85.5	104.1	112.9
FR	170.9	176.1	146.6	148.6	119.8	97.8	117.2	40.0
IT	—	—	113.2	115.6	106.8	74.2	90.6	147.5
ES	218.3	153.9	144.2	146.2	119.3	95.8	115.3	115.6

Notes: All estimates in percent of GDP. Ghosh et al. (2013): debt limits estimated from the interest payment schedule for 1985–2007. “Hist.”: average interest rate-growth differential for 1998–2007. “Proj.”: long-term government bond yield averaged over 2010–2014 (IMF projections as of 2010). Collard et al. (2015): maximum sustainable debt via a maximum primary surplus approach. “MPS”: maximum primary surplus; “MRR”: maximum recovery rate; “TVR”: time-varying interest rates; “CATA”: catastrophes; “h. MPS”: historical peak of the primary-surplus-to-GDP ratio.

Table 2: Parameter estimates under the baseline calibration

Parameter	Value	Parameter	Value
ρ_d	0.951	$\rho_{d,\ell}$	0.672
ρ_ℓ	0.999 [†]	$\bar{\ell}$	0.000 [†]
ρ_γ	0.781	ζ	1.500 [†]
ρ_i	0.981	$\sqrt{\text{Var}(i_t)}$	0.008 [†]
ρ_i^Q	0.999 [†]	$\max SR$	0.754
α	0.397	ν	0.449
$\bar{d}_{DE}$	0.706 [‡]	$\bar{d}_{FR}$	0.946 [‡]
$\bar{d}_{IT}$	1.285 [‡]	$\bar{d}_{ES}$	0.879 [‡]
$\bar{d}_{NL}$	0.585 [‡]	$\bar{d}_{BE}$	1.049 [‡]
$\bar{i}$	27.19 [‡]		

Notes: Parameter estimates under the baseline calibration. [†]: fixed during estimation. [‡]: calibrated directly from the data.

Table 3: Redistribution schemes and yield gains, $\theta = 0.05$

2008-06-30			2011-12-31			2021-06-30			
Redist. weight	Post-redist. yield	Yield gain	Redist. weight	Post-redist. yield	Yield gain	Redist. weight	Post-redist. yield	Yield gain	
SJG									
Panel A: Same funding costs, i.e. no ex-post redistribution									
DE	45%	594	−98	−37%	328	−75	45%	164	−100
FR	27%	594	−79	0%	328	−1	26%	164	−81
IT	6%	594	−22	87%	328	335	6%	164	−24
ES	11%	594	−67	47%	328	284	8%	164	−51
NL	8%	594	−74	−4%	328	−38	10%	164	−94
BE	4%	594	−63	7%	328	112	5%	164	−81
Panel B: Redistribution based on GDP weights									
DE	33%	567	−71	33%	190	64	33%	137	−73
FR	24%	586	−72	24%	261	66	24%	156	−74
IT	19%	645	−74	19%	585	78	19%	216	−76
ES	12%	599	−72	12%	536	76	12%	188	−75
NL	8%	591	−72	8%	225	65	8%	143	−73
BE	4%	603	−72	4%	370	70	4%	156	−74
Panel C: Same yield gains across countries									
DE	33%	568	−72	35%	185	69	33%	138	−74
FR	24%	586	−72	25%	258	69	24%	157	−74
IT	19%	644	−72	17%	595	69	19%	214	−74
ES	12%	599	−72	11%	543	69	12%	187	−74
NL	8%	591	−72	8%	221	69	8%	143	−74
BE	4%	602	−72	4%	372	69	4%	157	−74
SNJG									
Panel D: Same funding costs									
DE	−	522	−26	−	397	−143	−	90	−26
FR	−	522	−7	−	397	−70	−	90	−7
IT	−	522	50	−	397	266	−	90	50
ES	−	522	5	−	397	215	−	90	23
NL	−	522	−2	−	397	−107	−	90	−21
BE	−	522	9	−	397	44	−	90	−7

Notes: Yields and yield gains in basis points. “Redist. weight”: redistribution weight assigned to each country. “Post-redist. yield”: yield after redistribution. “Yield gain”: relative to the corresponding national yield. SJG: joint and several guarantees; SNJG: several-but-not-joint guarantees.

Table 4: Redistribution schemes and yield gains, $\theta = 0.10$

2008-06-30			2011-12-31			2021-06-30			
	Redist. weight	Post-redist. yield	Yield gain	Redist. weight	Post-redist. yield	Yield gain	Redist. weight	Post-redist. yield	Yield gain
SJG									
Panel A: Same funding costs, i.e. no ex-post redistribution									
DE	39%	621	−115	−48%	364	−80	41%	215	−119
FR	27%	621	−105	−9%	364	−20	27%	215	−105
IT	10%	621	−52	102%	364	321	9%	215	−45
ES	12%	621	−96	54%	364	266	9%	215	−73
NL	8%	621	−94	−6%	364	−45	9%	215	−115
BE	4%	621	−91	7%	364	92	5%	215	−107
Panel B: Redistribution based on GDP weights									
DE	33%	601	−95	33%	231	53	33%	191	−94
FR	24%	610	−95	24%	289	55	24%	205	−95
IT	19%	667	−98	19%	621	65	19%	268	−98
ES	12%	620	−96	12%	567	63	12%	239	−96
NL	8%	622	−96	8%	265	54	8%	195	−94
BE	4%	626	−96	4%	398	58	4%	203	−94
Panel C: Same yield gains across countries									
DE	33%	602	−96	35%	227	57	33%	192	−95
FR	24%	611	−96	25%	287	57	24%	206	−95
IT	19%	665	−96	17%	629	57	19%	265	−95
ES	12%	620	−96	11%	573	57	12%	238	−95
NL	8%	622	−96	8%	262	57	8%	196	−95
BE	4%	626	−96	4%	400	57	4%	203	−95
SNJG									
Panel D: Same funding costs									
DE	−	525	−19	−	420	−136	−	120	−23
FR	−	525	−10	−	420	−77	−	120	−10
IT	−	525	44	−	420	265	−	120	50
ES	−	525	−1	−	420	210	−	120	22
NL	−	525	2	−	420	−102	−	120	−19
BE	−	525	5	−	420	36	−	120	−12

Notes: Yields and yield gains in basis points. “Redist. weight”: redistribution weight assigned to each country. “Post-redist. yield”: yield after redistribution. “Yield gain”: relative to the corresponding national yield. SJG: joint and several guarantees; SNJG: several-but-not-joint guarantees.

Table 5: Redistribution schemes and yield gains, $\theta = 0.20$

2008-06-30			2011-12-31			2021-06-30			
Redist. weight	Post-redist. yield	Yield gain	Redist. weight	Post-redist. yield	Yield gain	Redist. weight	Post-redist. yield	Yield gain	
SJG									
Panel A: Same funding costs, i.e. no ex-post redistribution									
DE	6%	439	4	-1%	211	-2	-4%	-2	-3
FR	-5%	439	-5	1%	211	3	-2%	-2	-3
IT	90%	439	122	61%	211	439	85%	-2	128
ES	0%	439	0	32%	211	356	21%	-2	48
NL	8%	439	25	3%	211	41	1%	-2	3
BE	1%	439	6	5%	211	145	-1%	-2	-4
Panel B: Redistribution based on GDP weights									
DE	33%	418	25	33%	88	120	33%	-33	28
FR	24%	409	25	24%	93	121	24%	-32	28
IT	19%	534	26	19%	501	149	19%	97	29
ES	12%	414	25	12%	424	143	12%	18	28
NL	8%	439	25	8%	129	123	8%	-27	28
BE	4%	420	25	4%	227	129	4%	-33	28
Panel C: Same yield gains across countries									
DE	33%	418	25	35%	80	128	33%	-33	28
FR	24%	409	25	26%	86	128	24%	-32	28
IT	18%	535	25	16%	521	128	18%	98	28
ES	12%	414	25	11%	439	128	12%	18	28
NL	8%	439	25	8%	124	128	8%	-27	28
BE	4%	420	25	4%	228	128	4%	-34	28
SNJG									
Panel D: Same funding costs									
DE	—	464	-21	—	339	-130	—	26	-31
FR	—	464	-30	—	339	-125	—	26	-31
IT	—	464	96	—	339	311	—	26	100
ES	—	464	-25	—	339	228	—	26	20
NL	—	464	0	—	339	-87	—	26	-25
BE	—	464	-19	—	339	17	—	26	-32

Notes: Yields and yield gains in basis points. “Redist. weight”: redistribution weight assigned to each country. “Post-redist. yield”: yield after redistribution. “Yield gain”: relative to the corresponding national yield. SJG: joint and several guarantees; SNJG: several-but-not-joint guarantees.

Table 6: Redistribution schemes and yield gains, $\theta = 0.30$

2008-06-30			2011-12-31			2021-06-30			
	Redist. weight	Post-redist. yield	Yield gain	Redist. weight	Post-redist. yield	Yield gain	Redist. weight	Post-redist. yield	Yield gain
SJG									
Panel A: Same funding costs, i.e. no ex-post redistribution									
DE	3%	438	3	-4%	223	-16	2%	-9	2
FR	0%	438	0	2%	223	10	0%	-9	0
IT	77%	438	141	62%	223	441	78%	-9	146
ES	9%	438	26	32%	223	357	16%	-9	47
NL	9%	438	38	2%	223	37	3%	-9	14
BE	2%	438	16	5%	223	151	0%	-9	3
Panel B: Redistribution based on GDP weights									
DE	33%	408	34	33%	89	117	33%	-42	34
FR	24%	405	34	24%	114	118	24%	-43	34
IT	19%	543	36	19%	518	146	19%	100	37
ES	12%	430	34	12%	440	140	12%	3	35
NL	8%	442	34	8%	140	120	8%	-30	34
BE	4%	421	34	4%	247	127	4%	-40	34
Panel C: Same yield gains across countries									
DE	33%	407	34	35%	81	125	33%	-42	35
FR	24%	404	34	25%	108	125	24%	-44	35
IT	18%	545	34	16%	539	125	18%	102	35
ES	12%	430	34	11%	454	125	12%	3	35
NL	8%	443	34	8%	135	125	8%	-30	35
BE	4%	420	34	4%	248	125	4%	-41	35
SNJG									
Panel D: Same funding costs									
DE	—	472	-31	—	348	-142	—	25	-33
FR	—	472	-34	—	348	-115	—	25	-34
IT	—	472	107	—	348	316	—	25	111
ES	—	472	-8	—	348	232	—	25	12
NL	—	472	4	—	348	-88	—	25	-21
BE	—	472	-18	—	348	25	—	25	-31

Notes: Yields and yield gains in basis points. “Redist. weight”: redistribution weight assigned to each country. “Post-redist. yield”: yield after redistribution. “Yield gain”: relative to the corresponding national yield. SJG: joint and several guarantees; SNJG: several-but-not-joint guarantees.

Table 7: Redistribution schemes and yield gains, $\theta = 0.40$

2008-06-30			2011-12-31			2021-06-30			
	Redist. weight	Post-redist. yield	Yield gain	Redist. weight	Post-redist. yield	Yield gain	Redist. weight	Post-redist. yield	Yield gain
SJG									
Panel A: Same funding costs, i.e. no ex-post redistribution									
DE	5%	437	5	-4%	221	-14	3%	-10	3
FR	1%	437	2	3%	221	13	1%	-10	2
IT	74%	437	143	61%	221	444	76%	-10	147
ES	10%	437	29	32%	221	361	16%	-10	49
NL	9%	437	41	2%	221	40	3%	-10	15
BE	2%	437	19	5%	221	154	1%	-10	5
Panel B: Redistribution based on GDP weights									
DE	33%	406	36	33%	87	119	33%	-43	36
FR	24%	404	36	24%	113	121	24%	-44	36
IT	19%	542	38	19%	516	149	19%	98	38
ES	12%	430	36	12%	438	143	12%	2	36
NL	8%	441	36	8%	138	123	8%	-31	36
BE	4%	420	36	4%	246	130	4%	-41	36
Panel C: Same yield gains across countries									
DE	33%	406	36	35%	79	128	33%	-43	36
FR	24%	403	36	25%	106	128	24%	-45	36
IT	18%	543	36	16%	537	128	18%	100	36
ES	12%	430	36	11%	453	128	12%	2	36
NL	8%	441	36	8%	133	128	8%	-31	36
BE	4%	419	36	4%	247	128	4%	-41	36
SNJG									
Panel D: Same funding costs									
DE	—	473	-31	—	349	-142	—	26	-33
FR	—	473	-34	—	349	-115	—	26	-34
IT	—	473	106	—	349	316	—	26	111
ES	—	473	-7	—	349	233	—	26	13
NL	—	473	4	—	349	-88	—	26	-21
BE	—	473	-18	—	349	26	—	26	-31

Notes: Yields and yield gains in basis points. “Redist. weight”: redistribution weight assigned to each country. “Post-redist. yield”: yield after redistribution. “Yield gain”: relative to the corresponding national yield. SJG: joint and several guarantees; SNJG: several-but-not-joint guarantees.

Table 8: Redistribution schemes with protected funding costs, $\theta = 0.20$

	2008-06-30			2011-12-31			2021-06-30		
	Redist. weight	Post-redist. yield	Yield gain	Redist. weight	Post-redist. yield	Yield gain	Redist. weight	Post-redist. yield	Yield gain
Panel A: No change in German funding cost									
DE	0%	443	0	0%	208	0	0%	-5	0
FR	36%	397	37	36%	37	177	36%	-45	41
IT	28%	521	39	28%	432	217	28%	83	44
ES	18%	402	37	18%	358	209	18%	4	42
NL	11%	426	37	11%	72	180	11%	-40	41
BE	7%	408	37	7%	167	189	7%	-47	41
Panel B: No change in German and French funding cost									
DE	0%	443	0	0%	208	0	0%	-5	0
FR	0%	434	0	0%	214	0	0%	-4	0
IT	44%	500	61	44%	321	329	44%	59	68
ES	28%	382	57	28%	250	316	28%	-18	65
NL	18%	406	58	18%	-22	273	18%	-62	63
BE	10%	388	57	10%	69	287	10%	-69	63

Notes: Yields and yield gains in basis points. Panel A imposes zero change in Germany's funding cost. Panel B imposes zero change in both Germany's and France's funding costs. Redistribution weights are adjusted accordingly.